

## Henrique de Freitas Serra

### Full-Stack & Research Software Engineer

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### Professional Summary

Full-stack and research software engineer specializing in large-scale data systems, mobile platforms, and scientific software infrastructure. Experienced in designing and deploying full-stack applications integrating mobile apps, backend systems, and research data pipelines.

Currently the lead developer of **BDMobility**, a research platform processing more than **200,000 recorded trips and over 20 million geospatial coordinates**, used by researchers studying urban mobility and migration patterns.

My work focuses on building advanced computational tools for academic research, including geospatial analysis pipelines, distributed data processing systems, and interactive dashboards that allow researchers to explore complex datasets.

Before transitioning fully into software engineering, I completed four years of university-level physics training, developing strong analytical and mathematical foundations that continue to inform my engineering work.

My technical contributions and research collaboration led to an invitation to pursue doctoral research within an academic environment despite not having formally completed a bachelor's degree.

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### Core Technical Skills

#### Programming Languages

- **JavaScript / TypeScript — Expert**
- **Dart (Flutter) — Advanced**
- Java — Intermediate (research software maintenance and modernization)
- Kotlin — Android development

- Swift — iOS development
  - SQL
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### **Frameworks & Platforms**

- React
  - Next.js
  - Angular
  - Node.js
  - Express
  - Flutter (iOS, Android, Web)
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### **Databases**

- PostgreSQL (Neon serverless Postgres)
  - MongoDB
  - Firebase
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### **DevOps & Cloud**

- Docker
  - GitHub Actions
  - Vercel
  - AWS
  - Serverless architectures
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### **Specialized Engineering Areas**

- Geospatial data processing
- Trip clustering and route similarity analysis
- Fréchet distance path comparison
- Parallel processing with Web Workers

- Data pipelines for mobility datasets
  - Binary file parsing and reverse engineering
  - Interpreter and compiler design
  - Research software development
  - Scientific data visualization
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## Professional Experience

### Full-Stack Developer

#### Bridging Divides Research Initiative

Montreal, Canada | 2024 – Present

Lead developer of **BDMobility**, a research platform designed to analyze transportation behavior and urban mobility patterns.

Responsibilities and achievements include:

- Designed and implemented a **full-stack research platform** including Flutter mobile applications, backend services, and analytical dashboards.
- Built a geospatial data pipeline capable of processing **200,000+ recorded trips and more than 20 million GPS coordinates**.
- Implemented **trip clustering and route similarity algorithms** to detect recurring mobility patterns.
- Developed systems for identifying **traffic delays and route deviations**.
- Built research dashboards enabling investigators to explore datasets and mobility trends.
- Designed backend infrastructure using **Next.js, TypeScript, and Neon PostgreSQL**.
- Deployed and maintained production infrastructure using **Vercel, Docker, and cloud services**.
- Worked directly with interdisciplinary research teams to translate research questions into computational systems.

The platform is distributed through a **public Flutter mobile application available on app stores**.

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## Research Software Developer

### PTIDEJ Research Group – Concordia University

Montreal, Canada | 2024 – Present

Contributor to **PTIDEJ**, a major academic software engineering research platform used to analyze object-oriented software architecture.

Key contributions:

- Modernized core parsing infrastructure including **CFParse**, responsible for constant pool parsing and binary file interpretation.
- Updated legacy components to ensure compatibility with **modern Java versions**.
- Fixed numerous architectural and stability issues across the platform.
- Contributed improvements to research tools used for analyzing software design and patterns.

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## Major Engineering Projects

### University Conference Management Platform

Concordia University | 2025 – Present

Designing and implementing a **complete conference management infrastructure** used by student organizers and academic committees.

Key capabilities include:

- Full database, backend, and frontend architecture.
- Multi-role dashboards for organizers, presenters, judges, and sponsors.
- Tools for **creating and managing conference schedules** with automated suggestions and autofill.
- Payment systems and registration management.
- Submission and publication systems for **presentations and posters**.
- Judge assignment and automated scheduling tools.
- Sponsor dashboards for managing partnerships and event participation.

The system provides a **centralized platform for organizing large academic conferences**, replacing manual workflows with integrated digital tools.

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## BDMobility – Mobility Analysis Platform

Technologies: Flutter, Next.js, TypeScript, PostgreSQL, Docker

A full research platform designed for large-scale analysis of urban mobility behavior.

Key features:

- Mobile application collecting GPS mobility data.
- Backend pipeline processing **20M+ coordinates**.
- Route similarity detection and trip clustering.
- Traffic delay analysis tools.
- Interactive research dashboards.

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## CRVJA – AMOS BASIC Interpreter and Emulator

Technologies: JavaScript, React, Next.js

CRVJA is a browser-based interpreter for the **AMOS BASIC programming language** originally used on the Commodore Amiga.

Key features:

- Reverse-engineered **AMOS tokenized binary format**.
- Built a tokenizer and interpreter capable of executing legacy programs.
- Implemented browser-based rendering of graphics.
- Added support for **sprite banks and Amiga assets**.
- Used in **Concordia programming camps** to teach programming fundamentals.

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## AMOS Binary Decoder

Developed tools capable of decoding tokenized AMOS BASIC files and reconstructing readable source code.

Key features:

- Reverse engineering of binary token structures
- Parsing commands, variables, strings, and floating-point values
- Reconstruction of original program structure

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## Conference Presentations

Presented technical work and research tools at academic conferences including:

- **Reanimate School 2025**
- **Bridging Divides Discovery Research Conference**

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## Open Source Contributions

Contributor to research and educational software projects including:

- CRVJA – AMOS BASIC Interpreter
- PTIDEJ Software Engineering Platform

GitHub:

[github.com/agadfs](https://github.com/agadfs)

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## Education

### Computer Science

Estácio de Sá University – Brazil

2023 – 2025 (coursework completed)

### Physics

Federal University of Itajubá – Brazil

2019 – 2023

Four years of university-level physics studies with emphasis on analytical and mathematical problem solving.

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## Additional Technical Training

Completed extensive professional training through the **Alura technology platform**, equivalent to approximately **three years of full-time full-stack software development study**, covering modern frontend, backend, and software engineering practices.

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## Languages

Portuguese — Native

English — Fluent

French — Basic

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### **Research Contributions**

Second author on research work supporting academic publications and graduate-level research projects.